





Exam :

Subject :

Total Marks :

QP :

End Term Quiz

LLM

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2025 Aug31: IIT M AN EXAM QIA4

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QUESTION MENU

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Question 1 : 6406531431741

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**DEGREE LEVEL : LARGE LANGUAGE MODELS (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? **CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)**

OPTIONS :

☐ YES

☐ NO

[Discussions \(0\)](#)

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Question 2 : 6406531431742

[View Solutions \(0\)](#) Total Mark : 3.00 | Type : MCQ



Given the input string:



sunshine



And the following vocabulary of subword tokens with their corresponding log-probabilities:

Subword	Log-Probability (base e)
sun	-0.5
shine	-0.7
sunshine	-1.6
su	-0.3
nshine	-0.4
n	-1.8
shi	-0.6
ne	-0.5

Using the Viterbi algorithm (as used in the SentencePiece tokenizer), determine the most probable tokenization of the input string. The probability of a tokenized sequence is the sum of the log-probabilities of the selected subwords. Only subwords from the vocabulary may be used.

OPTIONS :

- ☐ [sunshine]
- ☐ [su, nshine]
- ☐ [sun, shine]
- ☐ [sun, shi, ne]

Discussions (0)

Suggest Correction

Question 3 : 6406531431743

View Solutions (0) Total Mark : 2.00 | Type : MCQ

A GPT model is trained using **causal language modeling**. During training, for a sequence of $T = 4$ tokens, which of the following correctly represents the **attention mask matrix** applied to the attention logits?

OPTIONS :

- ☐ $\begin{bmatrix} 0 & -\infty & -\infty & -\infty \\ 0 & 0 & -\infty & -\infty \\ 0 & 0 & 0 & -\infty \\ 0 & 0 & 0 & 0 \end{bmatrix}$
- ☐ $\begin{bmatrix} 0 & \infty & \infty & \infty \\ 0 & 0 & \infty & \infty \\ 0 & 0 & 0 & \infty \\ 0 & 0 & 0 & 0 \end{bmatrix}$
- ☐ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$
- ☐ $\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Discussions (0)

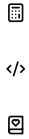
Suggest Correction

Broken Format?

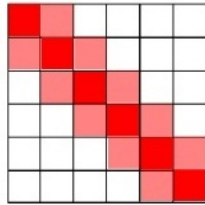
Question 4 : 6406531431744

View Solutions (0) Total Mark : 2.00 | Type : MCQ

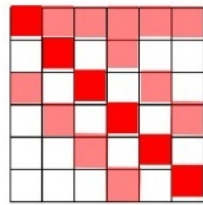
Total mark : 2.00 | Type : MSQ



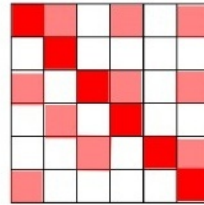
Match the given attention masks with their corresponding variants of attention mechanisms. In each figure, the colored cells indicate positions where a query token (Q_i) attends to a key token (K_j), while the white cells represent positions that are not attended. Choose the correct option.



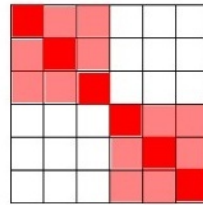
(a)



(b)



(c)



(d)

OPTIONS :

- ☐ (a) : Random Local Attention,(b) : Strided Local Attention, (c) : Sparse Block Attention, (d) : Local + Global Attention
- ☐ (a) : Strided Local Attention,(b) : Local + Global Attention, (c) : Random Local Attention, (d) : Sparse Block Attention
- ☐ (a) : Strided Local Attention,(b) : Random Local Attention, (c) : Local + Global Attention, (d) : Sparse Block Attention
- ☐ (a) : Sparse Block Attention,(b) : Local + Global Attention, (c) : Strided Local Attention, (d) : Random Local Attention

Discussions (0)

Suggest Correction

Question 5 : 6406531431745

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following preprocessing steps are commonly used when preparing data for transformer models like BERT?

OPTIONS :

- ☐ Adding special tokens such as [CLS] and [SEP] to the sequence
- ☐ Splitting tokens into subword units using a model-specific tokenizer
- ☐ Removing all punctuation marks to ensure cleaner embeddings
- ☐ Adding positional information to each token embedding

Discussions (0)

Suggest Correction

Broken Format?

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Question 6 : 6406531431746

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which modifications are used in transformers to handle longer sequences efficiently?

OPTIONS :

- ☐ Using sparse attention patterns to reduce computational complexity
- ☐ Applying low-rank matrix factorization to approximate attention
- ☐ Increasing only the feed-forward network width without modifying attention
- ☐ Replacing the softmax function in attention with other activation function for faster training

Discussions (0)

Suggest Correction

Broken Format?

Question 7 : 6406531431747

View Solutions (0) Total Mark : 3.00 | Type : MSQ

In Transformer models that use relative position embeddings (such as in Transformer-XL or T5), clipping is often applied to the relative position indices. Which of the following statements about clipping in relative position embeddings are correct?

OPTIONS :

- ☐ Clipping ensures that very large relative distances are mapped to a fixed maximum distance.
- ☐ Without clipping, the model would require embeddings for every possible relative distance, which is infeasible for long sequences.
- ☐ Clipping increases the precision of embeddings for large relative distances.
- ☐ Clipping introduces an upper bound k such that all distances greater than k are mapped to the same index.

Discussions (0)

Suggest Correction

Broken Format?

Question 8 : 6406531431748

View Solutions (0) Total Mark : 2.00 | Type : SA

Consider the following probability distribution over a vocabulary of 10 tokens:

[0.06, 0.20, 0.14, 0.06, 0.09, 0.08, 0.05, 0.15, 0.06, 0.11]

You apply **top- p (nucleus) sampling** with $p = 0.7$. How many tokens are included in the sampling pool?

Answer (Numeric):

Answer

Discussions (0)

Suggest Correction

Question 9 : 6406531431749

Total Mark : 0.00 | Type : COMPREHENSION

A vocabulary $\mathcal{V} = \{\text{tea, you, enjoy, often}\}$ is associated with the following embedding matrix:

$$E = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Words are indexed in the order shown above: “tea” (index 0), “you” (1), “enjoy” (2), and “often” (3). No positional encodings are used.

The parameters of the attention layer are:

$$W_Q = \begin{bmatrix} 0.5 & 1 \\ 1 & 0.5 \end{bmatrix}, \quad W_K = \begin{bmatrix} 1 & 0.5 \\ -0.5 & 1 \end{bmatrix}, \quad W_V = \begin{bmatrix} 1 & 0.2 \\ 0.5 & 1 \end{bmatrix},$$
$$W_O = \begin{bmatrix} 0.5 & 0.5 \\ 1 & -1 \end{bmatrix}$$

For the input sequence “you enjoy tea often”. The computed attention matrix $A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)$ is given as:

$$A = \begin{bmatrix} 0.55 & 0.32 & 0.11 & 0.02 \\ 0.43 & 0.25 & 0.21 & 0.1 \\ 0.39 & 0.39 & 0.16 & 0.07 \\ 0.24 & 0.4 & 0.2 & 0.17 \end{bmatrix}$$

Answer the given subquestions. If you believe the information is insufficient, enter -1 as your answer.

Question 10 : 6406531431750

View Report ON

View Solutions (0)

Question 10 : 6406531431750

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

For the input “you enjoy tea often”, compute the final representation of the word “enjoy” after the attention layer (i.e., after applying WQ, WK, WV , attention weights, and WO). Enter the sum of the elements in the resulting vector.

Answer (Numeric):

[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)

Question 11 : 6406531431751

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

Suppose the input sentence is “tea you enjoy often”. Using the same matrices and processing method, what is the attention score (i.e., softmax entry from A) for the query word “tea” attending to key word “often”?

Answer (Numeric):

[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)

Question 12 : 6406531431758

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Suppose we have context length $T = 30$ and the number of blocks $n = 5$ for the block attention mechanism.

The masking matrix M is defined as:

$$M_{ij} = \begin{cases} 1 & \text{if } \pi(\lfloor \frac{in}{T} \rfloor) = \lfloor \frac{jn}{T} \rfloor, \\ 0 & \text{otherwise} \end{cases}$$

where i is row indices and j is columns indexes both starting from 0.

Here, π is a permutation of $\{0, 1, \dots, (n-1)\}$, which determines the allowed query-key block pairs. Specifically, if a query token i lies in block

$$r = \lfloor \frac{in}{T} \rfloor,$$

then it can attend only to key tokens in block $\pi(r)$.

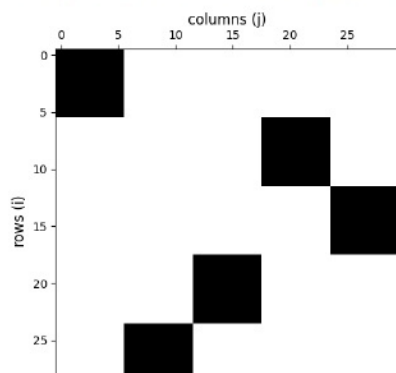
Equivalently, if we partition queries and keys into blocks,

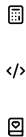
$$Q = (Q_0, Q_1, \dots, Q_{n-1}), \quad K = (K_{\pi(0)}, K_{\pi(1)}, \dots, K_{\pi(n-1)}),$$

then block Q_r attends only to block $K_{\pi(r)}$.

This block-restricted attention reduces computation by limiting interactions to the query-key block pairs specified by π .

Answer the given subquestions using the attention mask $M \in \{0, 1\}^{T \times T}$ image, where black squares represent entries equal to 1 and white squares represent entries equal to 0.





Broken Format?

Question 13 : 6406531431759

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

Choose the correct representation of π for the given mask image.

OPTIONS :

- ☐ $\pi = (2, 1, 3, 4, 0)$
- ☐ $\pi = (0, 4, 3, 1, 2)$
- ☐ $\pi = (0, 3, 4, 2, 1)$
- ☐ All of these

[Discussions \(0\)](#)

[Suggest Correction](#)

Broken Format?

Question 14 : 6406531431760

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

How many permutations of π are possible for $n = 5$?

Answer (Numeric):

Answer

[Discussions \(0\)](#)

[Suggest Correction](#)

Broken Format?

Question 15 : 6406531431761

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

What percentage of entries in the attention matrix are zero (i.e., the sparsity)?

Answer (Numeric):

Answer

[Discussions \(0\)](#)

[Suggest Correction](#)

Broken Format?

Question 16 : 6406531431752

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the short sentence:

“small models sometimes beat big ones”

A model processes this sequence using the *naive relative positional embedding* method. The model’s embedding dimension is $d_{\text{model}} = 4$, and the number of tokens is $T = 6$. We use **zero-based indexing** for token positions:

0 : small 1 : models 2 : sometimes 3 : beat 4 : big 5 : ones

Note: In this context, the *relative position* between two tokens is:

$$\text{Relative position} = j - i$$

(rows = current token i , columns = other token j).

1. Token embeddings (rows correspond to tokens in order):



$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 5 & 5 & 5 & 5 \\ 6 & 6 & 6 & 6 \end{bmatrix}.$$

Denote the i -th row by $\mathbf{x}_i$.

2. **Positional embedding definition:** For a relative position,

$$\mathbf{p}_{(j-i)} = \left[\frac{(j-i)}{5} \quad \frac{(j-i)}{5} \quad \frac{(j-i)}{5} \quad \frac{(j-i)}{5} \right].$$

[Broken Format?](#)

Question 17 : 6406531431753

[View Parent QN](#)
[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

Choose the sequence of *relative positions* $(j-i)$ for the token "sometimes" with respect to each token in the sentence.

OPTIONS :

- ☐ (0, 1, 2, 3, 4, 5)
- ☐ (-1, 0, 1, 2, 3, 4)
- ☐ (-3, -2, -1, 0, 1, 2)
- ☐ (-2, -1, 0, 1, 2, 3)

[Discussions \(0\)](#)
[Suggest Correction](#)

Question 18 : 6406531431754

[View Parent QN](#)
[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

Choose the value of $\mathbf{p}_{(j-i)}$ for the current token "big" and the other token "small".

OPTIONS :

- ☐ [-0.4,-0.4,-0.4,-0.4]
- ☐ [-0.1,-0.1,-0.1,-0.1]
- ☐ [-0.8,-0.8,-0.8,-0.8]
- ☐ [-0.2,-0.2,-0.2,-0.2]

[Discussions \(0\)](#)
[Suggest Correction](#)

Question 19 : 6406531431755

[View Parent QN](#)
[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

With a naive implementation, for each token embedding we combine (by addition) relative position embeddings.

For token index $i \in \{0, \dots, T-1\}$,

$$\mathbf{h}_i = \mathbf{x}_i + \sum_{i=0}^{T-1} \mathbf{p}_{j-i}$$



compute the final combined embedding $\mathbf{h}_i$ for the token "models" (i.e., $i = 1$).
Submit the sum of all the elements of $\mathbf{h}_1$

Answer (Numeric):

[Discussions \(0\)](#)[Suggest Correction](#)**Question 20 : 6406531431756**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

Compute e_{ij} for the current token “big” and other token “small”.

$$e_{ij} = x_i W_Q (x_j W_K + p_{(j-i)})^T$$

Assume $W_Q = I$ and $W_K = 2I$ where I is an identity matrix . Answer e_{ij} correct upto 1 digit after the decimal.

Answer (Numeric):

[Discussions \(0\)](#)[Suggest Correction](#)**Question 21 : 6406531431757**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

Now consider the same sequence but assume the model uses the *ALiBi* (*Attention with Linear Biases*) method. The model has $H = 4$ heads and the per-head slope is defined as

$$m_h = \frac{1}{2^h}, h = 0, 1, 2, 3,$$

The pre-attention e_{ij} for head h are computed as:

$$x_i W_Q x_j W_K + m_h \cdot [(j - i)]$$

Assume $W_Q = I$ and $W_K = 2I$ where I is an identity matrix . Answer e_{ij} for the current token “big” and other token “small” for head $h = 2$. correct upto 1 digit after the decimal.

Answer (Numeric):

[Discussions \(0\)](#)[Suggest Correction](#)**SUBMIT EXAM**